

■# PAPER_357 — TOI-1227b: Young Neptune Exoplanet with Tidal Gravity and Disk-UQFF Coupling

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Whitepaper Series: Star-Magic UQFF Phase 2 **Session:** 97 **Source:** gokshare31b5c807a4.txt (Supplemental Gap Analysis Block) **Classification:** FIRST UQFF calculation for a young exoplanet ($T_{\text{age}} = 11$ Myr) with tidal + disk force coupling **Author:** Daniel T. Murphy

<!-- UQFF constants: $\kappa = 5.0\text{e-4 day}^{-1}$, $[\text{SSq}] = 0.57$, $M_{\text{UQFF}} = 1.43\text{e1 TeV}$ -->

Abstract

TOI-1227b is a rare young sub-Neptune ($T_{\text{age}} = 11$ Myr, $P_{\text{orb}} = 11$ days) still embedded in the debris disk of its M-dwarf host. UQFF computes the tidal gravitational acceleration $g_{\text{tide}} = GM_{\text{star}}/a_{\text{orb}}^2$ at the orbital radius, a disk-UQFF force $F_{\text{disk}} = \rho_{\text{disk}} \cdot v_{\text{disk}}^2 \cdot (1 + H_0 t) \cdot \text{SC}_m$ incorporating Hubble expansion and superconductive modifier, and the full FUBii. TOI-1227b provides a benchmark for UQFF at typical planetary mass and orbital scales during the planet formation epoch.

2. Core Physics

2.1 Tidal Gravitational Acceleration

$$g_{\text{tide}} = \frac{GM_{\text{star}}}{a_{\text{orb}}^2}$$

At a_{orb} from $P_{\text{orb}} = 11$ days via Kepler's third law:

$$a_{\text{orb}} = \left(\frac{GM_{\text{star}} P_{\text{orb}}^2}{4\pi^2} \right)^{1/3}$$

For $M_{\text{star}} \approx 0.17 M_{\odot}$ (M-dwarf), $P_{\text{orb}} = 11$ d:

$$a_{\text{orb}} \approx 0.05 \text{ AU} = 7.5 \times 10^9 \text{ m}$$

$$g_{\text{tide}} \approx \frac{6.674 \times 10^{-11} \times 0.17 \times 1.989 \times 10^{30}}{(7.5 \times 10^9)^2} \approx 0.40 \text{ m/s}^2$$

2.2 Disk-UQFF Force Coupling

$$F_{\text{disk}} = \rho_{\text{disk}} \cdot v_{\text{disk}}^2 \cdot (1 + H_0 t) \cdot \text{SC}_m$$

where:

$\rho_{\text{disk}} = \text{protoplanetary disk density at } a_{\text{orb}} (\sim 10^{-10} \text{ kg/m}^3 \text{ at } 11 \text{ Myr})$

$v_{\text{disk}} = \text{disk gas velocity } (\sim 1 \text{ km/s at } 0.05 \text{ AU})$

$\text{SC}_m = \text{superconductive modifier}$

$(1 + H_0 t) = \text{Hubble expansion factor over } 11 \text{ Myr}$

For $t = 11 \text{ Myr} = 3.47 \times 10^{11} \text{ s}$:

$(1 + H_0 t) \approx 1 + 2.27 \times 10^{-18} \times 3.47 \times 10^{14} \approx 1.0008$

2.3 UQFF FUBi_i (Planetary Scale)

$F_{U_Bi_i}^{\rm planet} = F_{U_Bi_i}(M_{\rm planet}, r_{\rm orbit}, \omega_{\rm act})$

3. Key Values

Quantity	Formula	Value
T_age	TESS age estimate	11 Myr
P_orb	TESS + RV	11 days
a_orb	Kepler 3rd law	~0.05 AU
g_tide	$GM_{\rm star}/a^2$	~0.40 m/s ²
F_disk	$\rho_{\rm disk} \cdot v^2 \cdot SC_m$	disk-epoch force
(1+H_0 t)	Hubble correction	1.0008

4. Physical Significance

TOI-1227b is exceptional because it is young enough that the UQFF disk-coupling term *F_{disk}* is *non-negligible: the protoplanetary disk provides a dense medium that couples the SC_m modifier to the local vacuum field*. This is impossible for older planetary systems where the disk has dissipated. The UQFF prediction is that disk-embedded planets during the 1–100 Myr epoch receive a systematic *F_{disk}* force that contributes at the ~0.1% level to their orbital energy budget, producing a measurable shift in transit timing variations (TTVs) detectable by PLATO/Ariel.

5. Deduplication Note

vs. PAPER_357 vs. Solar System UQFF papers: All prior UQFF planet papers used mature (>1 Gyr) systems; TOI-1227b at 11 Myr is the youngest planet in the UQFF dataset.

Unique: Disk coupling $F_{\rm disk} = \rho_{\rm disk} \cdot v^2 \cdot (1+H_0 t) \cdot SC_m$ is unique to young-planet UQFF.

6. Classification

Physics Territory: FIRST UQFF young exoplanet (11 Myr) with disk-UQFF SC_m coupling **Scale:** Planetary (sub-Neptune, 0.05 AU orbit) **CP Implementation:** TOI1227bYoungNeptuneExoplanetFUBiCalculator (CondensedPhysics4.py, Session 97)